1.3.4 ▶ ▶ ▶ ▶ Relative Velocity for Elastic Collisions

When solving for the velocities in an elastic collision, we use two conservation laws: conservation of momentum, and since its an elastic collision, conservation of kinetic energy. These two conservation laws look like the following

$$m_1\mathbf{v}_{1i} + m_2\mathbf{v}_{2i} = m_1\mathbf{v}_{1f} + m_2\mathbf{v}_{2f} \tag{1.48}$$

$$\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 = \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2 \tag{1.49}$$

Because of the kinetic energy conservation equation, this system often gives a quadratic, which can often be cumbersome to solve. If we can find another equation that is linear, it would make the process of solving for the velocities of the particles much easier. Luckily, there is another equation that we can derive to use to make solving for the velocity easier. To derive this equation, we first rearrange both of the above equations to factor out the masses of the objects.

$$m_1(\mathbf{v}_{1i} - \mathbf{v}_{1f}) = m_2(\mathbf{v}_{2f} - \mathbf{v}_{2i})$$

$$m_1(v_{1i}^2 - v_{1f}^2) = m_2(v_{2f}^2 - v_{2i}^2)$$

Dividing the equations yields:

$$\boxed{\mathbf{v}_{1i} + \mathbf{v}_{1f} = \mathbf{v}_{2i} + \mathbf{v}_{2f}} \tag{1.50}$$

We can use this equation along with the conservation of momentum equation (1.47) to quickly solve for velocities in an elastic collision ★